

Periplaneta americana - type study

7.1 HABITAT AND HABIT

- 7.1 Habitat and habit
- 7.2 External features (Morphology)
- 7.3 Locomotion
- 7.4 Digestive system
- 7.5 Circulatory system
- 7.6 Respiratory system
- 7.7 Excretory system
- 7.8 Nervous system and sense organs
- 7.9 Reproductive system

Periplaneta americana, was a native of the tropical America. Now it has become a *cosmopolitan* insect. The name *Periplaneta americana* was coined by **Burmeister**. It is usually found in kitchens, hotels, bakeries, warehouses, etc., It is a **nocturnal** insect (it comes out of its hiding places to feed at night times). During the day time it remains inactive and takes rest. Its flattened form of the body enables it to enter small crevices in the floor or walls. It is an **omnivorous** insect, eating all types of food. It locates its food by the sense of smell. *Periplaneta* is a **cursorial** (swift runner) animal.

Classification

Phylum - Arthropoda
Class - Insecta
Order - Orthoptera
Family - Blattidae
Genus - *Periplaneta*
Species - *Americana*

Why do cockroaches take rest during day time?

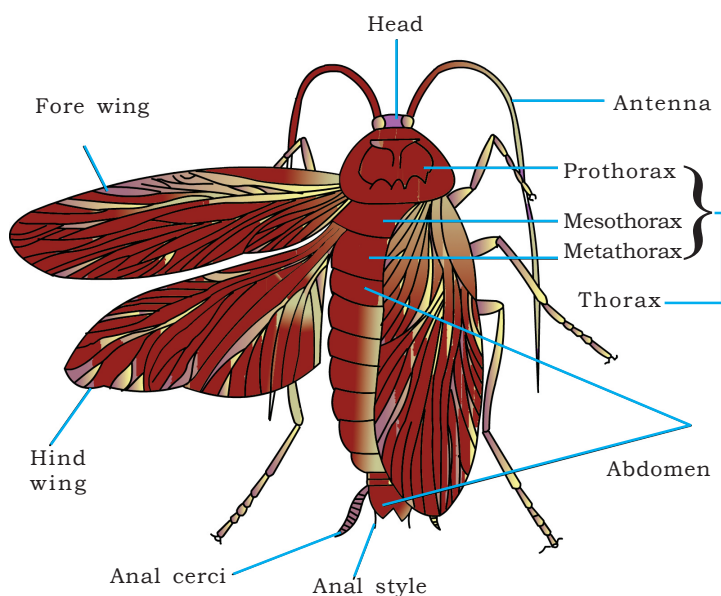


Fig. 7.1 *Periplaneta* (Male): External features

7.2 EXTERNAL FEATURES (MORPHOLOGY)

Cockroach has an elongated, narrow, elliptical, dorso-ventrally depressed body. It exhibits bilateral symmetry. The wings extend beyond the tip of the abdomen in the males. The colour is reddish brown with a light yellow area around the edges of the pronotum. The complete body is externally covered by a **chitinous cuticle** which constitutes the **exoskeleton**. The cuticle is secreted by the underlying cells of the

hypodermis^[1]. It is coated externally by a thin layer of wax, impermeable to water. The **oenocytes**, are believed to secrete **wax**. The cuticle protects the body, prevents loss of water, provides rigidity and offers place for the attachment of muscles. It consists of small plates or **sclerites**, which are joined by soft, flexible **arthrodial membranes**.

1. Body Divisions: The body of cockroach is distinctly segmented and consists of three **tagmata** (morphologically distinct parts of the body) namely head, thorax and abdomen.

A. Head: The head of cockroach is small and triangular. It is called **hypognathous** because it lies hanging almost at right angles to the body with the posterior wider part upwards, and the mouthparts directed down-wards.

The head of cockroach is formed by the fusion of six embryonic segments. It is movably attached to the thorax by a short neck or **cervicum**. It is covered by a number of sclerites which fuse to form a capsule. The top of the head between the eyes is called **vertex**. The vertex has two sclerites called '**epicranial plates**' connected by an '**epicranial suture**'. Below the vertex, the sclerites covering the head in front are – a large **frons**, a narrow rectangular **clypeus** and a movable **labrum**. Covering the sides of the head, below the compound eyes are the 'cheek sclerites' or '**genae**'. At the back of the head capsule there is a large opening called **occipital foramen**. It is bordered by a sclerite called **occiput**. The occipital foramen forms a passage for the oesophagus, aorta, nerve cord and tracheae. At the base of each **antenna**, a small whitish speck called **fenestra** or 'ocellar spot' or 'simple eye' is present. Appendages are absent in the first and third segments of the head. The second segment bears a pair of long, slender and segmented **antennae**, one on each side of the head. The antennae are tactile and olfactory

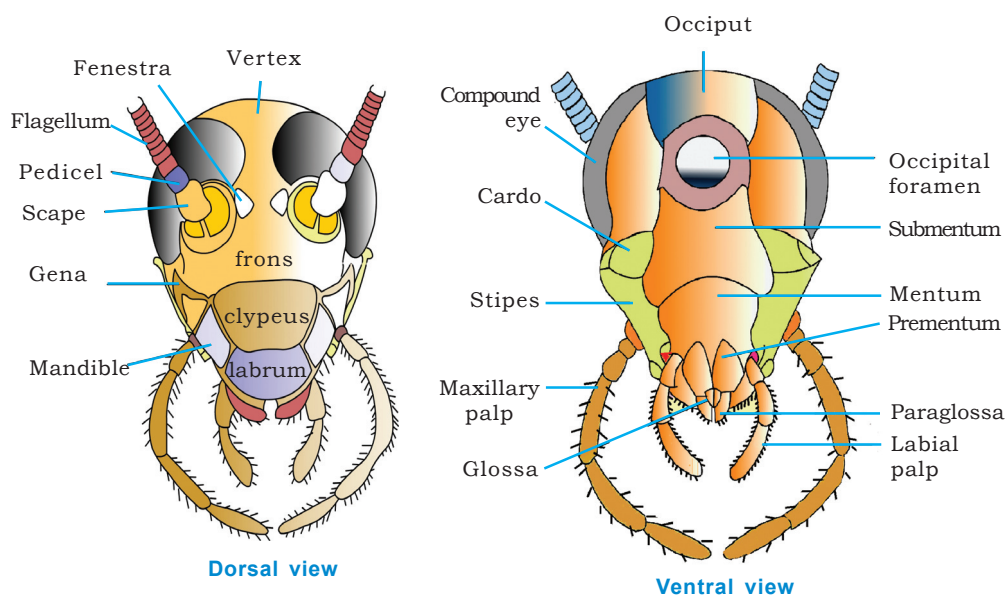


Fig. 7.2 *Periplaneta*: Head dorsal and ventral View

^[1] It is the epidermis below the cuticle, hence called hypodermis

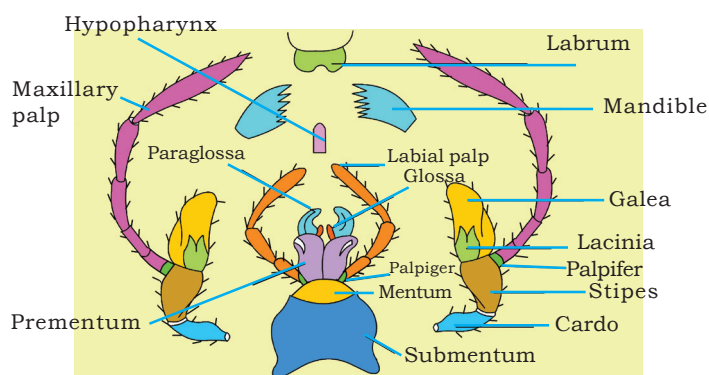


Fig. 7.3: *Periplaneta*: Mouth parts

in function. The fourth segment bears a pair of **mandibles**. The fifth segment has a pair of '**first maxillae**'. The sixth segment bears a pair of '**second maxillae**', which fuse to form the **labium** (also called 'lower lip').

Mouthparts

The mouthparts of *Periplaneta* are of **Biting** and **Chewing** type. These are the movably articulated appendages surrounding the mouth.

They include a pair of mandibles, a pair of first maxillae, a pair of second maxillae or labium and a tongue or **hypopharynx** and an upperlip, the labrum. They enclose a space called the '**preoral cavity**'.

Labrum (upper lip)

The labrum is a vertical, rectangular plate, movably connected with the clypeus. It forms the anterior wall of the preoral cavity and is also called the 'upper lip'. It bears '**gustatory sensillae**' on its inner surface. The labrum serves to hold the food and helps in tasting it.

Mandibles

A pair of triangular, hard, unjointed, chitinised mandibles is present on the sides of the mouth, connected with the genae. The inner margins of mandibles have teeth like structures. These teeth help to masticate the food. Two pairs of muscles called **adductor** and **abductor** muscles help in the movement of the mandibles. Each mandible has incising and grinding teeth.

First maxillae

A pair of first maxillae is situated on the sides of the mouth immediately behind the mandibles. Each maxilla is 'biramous' and consists of three parts: the protopodite, endopodite, and the exopodite. **Cardo** and **stipes** constitute the protopodite. A five jointed maxillary palp, arising from a sclerite called **palpifer**, on the outer side of the stipes, constitutes the exopodite. The maxillary palps are used for cleaning the antennae and the front pair of legs. From the distal end of the stipes, internal to the maxillary palp, arises the endopodite. Each endopodite has two parts, an outer broad hood-like **galea** and inner pincer-like (resembling a forceps) **lacinia**. Maxillae serve to hold the food and bring it to the mandibles.

Second maxillae (Labium or Lower lip)

The second pair of maxillae lies behind the first pair. Both the second maxillae fuse to form a broad plate like **labium** or **lower lip**. It consists of three parts: the upper

submentum, the middle mentum and the lower prementum. A three segmented labial palp, arising from a sclerite called **palpiger**, is present one on each side of the prementum. It prevents the food from falling, and pushes it into the preoral cavity.

The distal end of the prementum bears a pair of '**paraglossae** (comparable to **galeae** of the first maxilla), a pair of '**glossae**' (comparable to **laciniae** of the first maxilla). The glossae and paraglossae together constitute the **ligula**.

Hypopharynx (Tongue or Lingua): The hypopharynx is a rod like, grooved chitinous structure, hanging in the preoral cavity between the first maxillae. The hypopharynx is also called 'tongue' or '**lingua**'. It bears the opening of the '*efferent salivary duct*'.

Note: Biting and chewing type of mouth parts are believed to be the most primitive type (as most larvae of insects possess such type of mouth parts).

Why are the mouthparts of cockroach called biting and chewing type?

Neck (cervicum)

Neck is short, slender, flexible tube (a part or extension of the thorax) that connects the head with the thorax. It is supported by **four cervical sclerites**. It is not a 'tagma' of the body. It helps in the movement of the head.

B. Thorax: The thorax is the second 'tagma' of the body. It consists of three segments: prothorax, mesothorax, and metathorax. Each segment of the thorax is covered by four sclerites -one on the dorsal side (called **tergum** or **notum**), one on the ventral side (called **sternum**) and two (called **pleura**), on the lateral sides. The tergum of the prothorax is called **pronotum**. It is the largest sclerite of the body and covers the neck and a little part of the head. The terga on the mesothorax and metathorax are called **mesonotum** and **metanotum** respectively. The sclerites of each segment are joined by thin, soft, flexible membranes called arthrodial membranes, which permit movement, between the segments.

Legs: There are three pairs of jointed, walking legs in cockroach, attached one pair to each thoracic segment on the ventral side. The legs are connected with the pleura and sterna of the thoracic segments. According to the segment that bears them, they are called prolegs, mesolegs and metalegs. All the legs are similar in structure. Each leg is made up of five segments or **podomeres**. The different podomeres (from the base to the tip) are serially: **coxa**, **trochanter**, **femur**, **tibia** and **tarsus**.

The coxa connects the leg with the thorax. The trochanter is *movably attached* to the coxa, and *fused* with the femur. The femur and tibia bear chitinous bristles. The joints of tarsus are **tarsomeres** which bear soft pads called **plantulae** on their inner surface. The terminal joint of the tarsus, ends in a pair of **claws**. Between the claws there is a soft hairy pad called **pulvillus** or **arolium**.

Wings: Two pairs of wings, the fore wings and the hind wings, are present in cockroach. The forewings are thick, opaque and leathery. They do not help in flight, but cover and protect the hind wings when they are not in use. They are called **tegmina** (singular: **tegmen**). The hind wings are broad, thin, transparent and delicate.

They help in flight and remain folded below the tegmina when not in use. The wings of cockroach contain a network of hollow **veins** or **nervures**.

C. Abdomen: The abdomen consists of **ten** segments. Each segment is covered by the dorsal tergum, the ventral sternum and the two lateral pleura or pleurites. There are ten terga but only nine sterna as the tenth sternum is absent. The eighth tergum in the male and both eighth and ninth terga in the female are not visible as they are overlapped by the seventh tergum. The tenth tergum extends beyond the posterior end of the body and has a deep notch / groove in the middle of its free end. In the male nine sterna are visible whereas in the female, only seven sterna are visible. The seventh, eighth and ninth sterna together form a **brood pouch**. The brood pouch has two parts the anterior **genital chamber** or **gynatrium** and posterior **oothecal chamber**.

The posterior end of the abdomen has a pair of **anal cerci**, a pair of **anal styles** and gonapophyses in the males. Anal cerci are jointed (15 segmented) and arise from the lateral sides of the tenth tergum and are found in both the sexes. The anal styles are without joints and arise from the ninth sternum (seen only in the males). The **gonapophyses** are small chitinous processes arising from the ninth sternum in the males and eighth and ninth sterna in the females. They are the external genital organs. The **anus** is at the posterior end of the abdomen in the 10th segment. The genital aperture in male is present just below the anus on one of the **gonapophyses** and in female it is located on the **eighth sternum**.

Body wall: The body wall consists of three layers: **cuticle**, **epidermis** or **hypodermis** and **basement membrane**. The cuticle is the outer most layer. It is again differentiated into three layers, namely, an outer thin, waxy **epicuticle**, middle thick **exocuticle** made of tough pigmented chitin and an inner much thick **endocuticle**, made of soft layers of **chitin**. The articular membranes lack the exocuticle and so they are flexible. The epidermis lies below the cuticle. It comprises a single layer of columnar cells resting on the basement membrane. It secretes the cuticle. The basement membrane is the inner most layer of the body wall. The cuticle protects the delicate internal organs and prevents the loss of water by evaporation and provides surface for the attachment of muscles.

How does the cuticle help in preventing the loss of water?
What makes the articular membrane flexible?

Body cavity and coelom

The body cavity / perivisceral cavity of *Periplaneta* is not a true coelom and it is not lined by peritoneum. It is called '**haemocoel**' as it is filled with blood or haemolymph. The schizocoelom is restricted to the spaces around the reproductive organs.